ASSIGNMENT 03

1. **Accept n numbers from users and store them in an array. Write a function to sort and display sorted array.**
   i/p:[1,4,32,5]
   o/p:[1,4,5,32]

Code:-

```
#include<iostream>
using namespace std;
void acceptArray(int *arr, int size){
	cout<<"Enter Array elements: "<<endl;
	for(int i=0; i<size; i++)
	{
		cin>>arr[i];
	}
}

void displayArray(int *arr, int size){
	cout<<"Displaying Array: "<<endl;
	for(int i=0; i<size; i++)
	{
		cout<<arr[i]<<", ";
	}
	cout<<endl;
}

void sortArray(int *arr, int size){
	int tmp;
	for(int j=0; j<size;j++)
	{
		for(int i=0; i<size-1; i++){
			if(arr[i]>arr[i+1])
			{
				tmp = arr[i+1];
				arr[i+1] = arr[i];
				arr[i] = tmp;
			}
		}
	}
```

```
}

int main(){
        int size;
        cout<<"Enter the size of array: "<<endl;
        cin>>size;
        int arr[size];
        acceptArray(arr, size);
        displayArray(arr, size);
        sortArray(arr, size);
        displayArray(arr, size);
        return 0;
}
```

O/P:-

```
pgcp-ac@dac:~/Batch2/Assignment 03$ ./a.out
Enter the size of array:
5
Enter Array elements:
32
45
11
08
65
Displaying Array:
32, 45, 11, 8, 65,
Displaying Array:
8, 11, 32, 45, 65,
pgcp-ac@dac:~/Batch2/Assignment 03$
```

2. **Accept n characters from user and search a character appears how many times in the array.**
   i/p:['a','b','c','d','a']
   search 'a'
   o/p: a count is 2

Code:-

```
#include<iostream>
using namespace std;

void acceptdata(char *arr, int size){
	cout<<"Enter characters in array: "<<endl;
	for(int i=0; i<size; i++){
		cin>>arr[i];
	}
}

void displaydata(char *arr, int size){
	cout<<"[";
	for(int i=0; i<size; i++){
		cout<<"\'"<<arr[i]<<"\',";
	}
	cout<<"]"<<endl;
}

int searchdata(char *arr, int size, char ch){
	int count=0;
	for(int i=0; i<size; i++){
		if(arr[i]==ch){
			count++;
		}
	}
	return count;
}

int main(){
	int no;
	char c;
	cout<<"Enter no. of characters: "<<endl;
```

```
        cin>>no;
        char arr[no];
        acceptdata(arr, no);
        displaydata(arr, no);
        cout<<"Enter a character to search: ";
        cin>>c;
        cout<<c<<" count is "<<searchdata(arr, no, c)<<endl;
        return 0;
}
```

O/P:-

```
pgcp-ac@dac:~/Batch2/cpp day 3$ ./searchchar.out
Enter no. of characters:
5
Enter characters in array:
a
b
c
d
a
['a','b','c','d','a',]
Enter a character to search: a
a count is 2
pgcp-ac@dac:~/Batch2/cpp day 3$
```

3. **Accept n numbers from user and write a function to find out average and display average.**
   i/p:{10,20,30}
   o/p:Average: 20

Code:-

```
#include<iostream>
using namespace std;
void acceptdata(int *arr, int size){
	cout<<"Enter array elements: "<<endl;
	for(int i=0; i<size; i++){
		cin>>arr[i];
	}
}

void displaydata(int *arr, int size){
	cout<<"[";
	for(int i=0; i<size; i++){
		cout<<arr[i]<<",";
	}
	cout<<"]"<<endl;
}
int average(int *arr, int size){
	int avg=0;
	for(int i=0; i<size; i++){
		avg=(arr[0]+arr[size-1])/2;
	}
	return avg;
}
int main(){
	int n;
	cout<<"Enter total no. of elements: "<<endl;
	cin>>n;
	int arr[n];
	acceptdata(arr, n);
	displaydata(arr, n);
	cout<<"Average of all elements is: "<<average(arr, n)<<endl;
	return 0;
}
```

O/P:-

```
pgcp-ac@dac:~/Batch2/cpp day 3$ ./average.out
Enter total no. of elements:
3
Enter array elements:
10
20
30
[10,20,30,]
Average of all elements is: 20
pgcp-ac@dac:~/Batch2/cpp day 3$
```

4. **Accept n numbers from the user and write a function to multiply each num-ber by 2 and display the result.**
   i/p: [10,20,30,40]
   o/p:[20,40,60,80]

Code:-

```
	#include<iostream>
using namespace std;

void acceptdata(int *arr, int size){
	cout<<"Enter array elements: "<<endl;
	for(int i=0; i<size; i++){
		cin>>arr[i];
	}
}

void displaydata(int *arr, int size){
	cout<<"[";
	for(int i=0; i<size;i++){
		cout<<arr[i]<<",";
	}
	cout<<"]"<<endl;
}

void arraymultiplication(int *arr, int size){
	for(int i=0;i<size; i++){
		arr[i]=arr[i]*2;
	}
}

int main(){
	int n;
	cout<<"Enter no. of elements: "<<endl;
	cin>>n;
	int arr[n];
	acceptdata(arr, n);
	cout<<"Array elements before multiplication: "<<endl;
	displaydata(arr, n);
	arraymultiplication(arr, n);
```

```
        cout<<"\nArray elements after multiplication: "<<endl;
        displaydata(arr, n);
        return 0;
}
```

O/P:-

```
pgcp-ac@dac:~/Batch2/cpp day 3$ ./arraymultiply.out
Enter no. of elements:
5
Enter array elements:
10
20
30
40
50
Array elements before multiplication:
[10,20,30,40,50,]

Array elements after multiplication:
[20,40,60,80,100,]
pgcp-ac@dac:~/Batch2/cpp day 3$ S
```

5. **Accept n number from user and write functions for finding min, max, second max and second min from these numbers.**
   i/p: [10,20,30,40,5,7,10]
   o/p:
   min:10
   second min: 20
   max:40
   second max:30

Code:-

```
#include<iostream>
#include<climits>
using namespace std;
void acceptArray(int *arr, int size){
	cout<<"Enter Array elements: "<<endl;
	for(int i=0; i<size; i++)
	{
		cin>>arr[i];
	}
}

int findMin(int *arr, int size){
	int res=INT_MAX;
	for(int i=0; i<size; i++)
	{
		if(arr[i]<=res)
		{
			res=arr[i];
		}
	}
	return res;
}

int findMax(int *arr, int size){
	int res=INT_MIN;
	for(int i=0; i<size; i++)
	{
		if(arr[i]>=res)
		{
```

```
                res=arr[i];
            }
        }
        return res;
}

int findSecMin(int *arr, int size){
        int smin=INT_MAX;
        int min=INT_MAX;

        if(sizeof(arr) < 2){      return -1; }      //edge case

        for(int i=0; i<size; i++)                   //10,33,45,22,67
        {
            if(arr[i]<min)
            {
                smin = min;
                min=arr[i];
            }
            else if(arr[i]<smin && arr[i]!=min)
            {
                smin = arr[i];
            }
        }
        return smin;
}

int findSecMax(int *arr, int size){
        int smax=INT_MIN;
        int max=INT_MIN;

        if(sizeof(arr) < 2){      return -1; }  //edge case

        for(int i=0; i<size; i++)                   //10,33,45,22,67
        {
            if(arr[i]>max)
            {
                smax = max;
                max=arr[i];
            }
```

```
		else if(arr[i]>smax && arr[i]!=max)
		{
			smax = arr[i];
		}
	}
	return smax;
}

int main(){
	int size;
	cout<<"Enter the size of array: "<<endl;
	cin>>size;
	int arr[size];
	acceptArray(arr, size);
	int min = findMin(arr, size);		//Minimum Number
	cout<<"Minimum Number in array is: "<<min<<endl;

	int max = findMax(arr, size);		//Maximum Number
	cout<<"Maximum Number in array is: "<<max<<endl;

	int secMin = findSecMin(arr, size);		//Second Minimum Number
	cout<<"Second Minimum Number in array is: "<<secMin<<endl;

	int secMax = findSecMax(arr, size);		//Second Maximum Number
	cout<<"Second Maximum Number in array is: "<<secMax<<endl;
	return 0;
}
```

O/P:-

```
pgcp-ac@dac:~/Batch2/cpp day 3$ ./minMax.out
Enter the size of array:
5
Enter Array elements:
11
22
33
44
55
Minimum Number in array is: 11
Maximum Number in array is: 55
Second Minimum Number in array is: 22
Second Maximum Number in array is: 44
pgcp-ac@dac:~/Batch2/cpp day 3$
```